



Ahmedabad
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Milestone 1

Cross-Environment and SNR Robustness in Wi-Fi CSI-Based Human Activity Recognition: A Benchmark Study and Lightweight Domain-Shift Analysis

Category

AI Algorithms + Existing Dataset

Group 7

Name	Enrollment No.	Role
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Repository

github.com/Pranavv26/wifi-csi-domain-robustness

Abstract

WiFi Channel State Information (CSI) enables device free human activity recognition (HAR), but models trained in one setting degrade when the deployment domain changes, because CSI encodes the surrounding multipath channel as strongly as the target activity. This project addresses cross domain WiFi HAR and adopts, as its technical foundation, Domain Adversarial Test Time Adaptation (DATTA) [1], evaluated on the Widar3.0-G6D benchmark. DATTA combines domain adversarial training, test time adaptation, and random weight resetting on a lightweight WiFlexFormer backbone. Our project is scoped as a faithful reproduction, controlled evaluation, and component wise analysis of DATTA rather than a new algorithm. This Milestone 1 report defines the problem, develops a literature progression that motivates DATTA, characterizes Widar3.0-G6D and its Train/Val/Val_TTA/Test protocol, states the relevant state of the art position with appropriate qualification, and sets a staged implementation and evaluation plan. No experimental results are claimed for our team at this stage; all quantitative figures are attributed to the DATTA paper [1].

1. Introduction

WiFi sensing offers a low cost, unobtrusive, privacy preserving alternative to camera and wearable based HAR, using hardware already deployed indoors. Its central limitation is cross domain generalization: a model trained in one environment or on one set of users frequently fails elsewhere. This report scopes our team's project as a reproduction and controlled evaluation of a recent framework that treats real time cross domain robustness as its objective DATTA [1] on the dataset used in that work, Widar3.0-G6D. The report follows a single technical progression: from the CSI sensing mechanism, to why conventional HAR fails across domains, through the families of existing solutions and their residual limitations, to the specific gap that motivates DATTA, and finally to what our team will implement and measure.

2. Problem Definition and Motivation

A WiFi transmitter and receiver exchange OFDM signals across many subcarriers; CSI reports, per subcarrier, the amplitude attenuation and phase shift the channel imposes. Human motion perturbs multipath propagation, so distinctive CSI patterns correlate with activities and support HAR. The same dependence makes CSI a function of the propagation environment: room geometry, furniture, device placement, hardware, and interference shape CSI alongside the activity.

Following [1], a domain denotes the set of factors that shift the CSI distribution independently of the activity principally the physical environment and the individual, and more broadly the sensing hardware. Conventional HAR models minimize training domain loss and therefore encode domain specific channel signatures; when the domain changes at deployment, these signatures change and the learned decision boundaries no longer hold. WiFi CSI shifts are more abrupt than the smooth shifts typical in vision: a furniture rearrangement or a device relocation can move the distribution substantially [1].

Significance and motivation. Cross domain performance is the property that distinguishes a deployable system from a laboratory demonstration: a monitoring product is installed in an unseen home, possibly on different hardware, and must work immediately. This is a wireless communications problem a consequence of multipath propagation and channel variability studied in ECE 310 not solely a machine learning artifact.

Scope. Our project will reproduce the DATTA baseline and its components on Widar3.0-G6D, quantify the cross domain generalization gap, and analyze each component's contribution and the accuracy cost trade off. We will not collect new CSI data, design a new architecture, or claim state of the art beyond what a careful reproduction supports. We assume access to the public Widar3.0-G6D data and the authors' released code and subset preparation script [1]; where compute limits full reproduction we will reduce scale rather than alter the experimental design.

3. WiFi CSI Based HAR Background

A CSI measurement is a complex matrix over subcarriers and time (and antennas); its amplitude and phase across subcarriers form a time frequency representation from which activity discriminative features are learned. Classical pipelines use handcrafted statistical features; modern pipelines learn features with convolutional or transformer models directly on CSI amplitude, and phase where usable. The learned representation determines cross domain behaviour: a representation that captures activity relevant structure transfers to new domains, whereas one that captures environment specific channel structure does not. Domain invariant feature learning therefore seeks to retain the former while discarding the latter.

4. The Cross Domain Generalization Problem

Training and test data are drawn from different distributions because they originate in different domains. Two remedy families exist. Domain generalization removes domain specific statistics during training so the model is robust to unseen domains. Adaptation uses target information: domain adaptation aligns source and target during training, while test time adaptation (TTA) adjusts the model during inference on the unlabeled incoming stream. Each carries a cost: invariance alone can leave residual shift; training time adaptation presupposes target data; TTA risks instability and catastrophic forgetting. This tension motivates combining invariance with online adaptation under an explicit stability control, which is the design point DATTA occupies [1].

5. Related Work

The DATTA paper organizes cross domain WiFi HAR into physics informed representations, synthetic augmentation, adaptive learning, and large scale data diversity [1]. We summarize the strands most relevant to our direction as a progression rather than a catalogue; for each we state the problem addressed, the core idea, the improvement, and the residual limitation.

Table 1: Relevant strands in cross domain WiFi HAR and their relationship to DATTA.

Strand	Core idea / what it improves	Residual limitation → relevance to DATTA
Physics informed / domain invariant representations	Exploit antenna phase differences or body coordinate velocity profiles, and filter static interference, to reduce sensitivity to the environment; improves transfer without target data	Handcrafted invariance is partial and representation specific; residual domain shift persists, motivating learned invariance.
Environment independent WiFi HAR via domain adversarial learning	Train a feature extractor adversarially against a domain discriminator so features are environment agnostic; single classifier WiFi variants keep inference cheap [3,4]	Adversarial invariance addresses static gaps but not the rapid, unpredictable shifts seen at deployment; residual shift remains, motivating adaptation.
CSI specific augmentation	Apply realistic CSI perturbations (amplitude perturbation, temporal circular rotation, pixel/row operations) to increase variability; substantially strengthens adversarial training [1]	Gains are augmentation and dataset dependent and insufficient alone; used as an enabling component within DATTA.
Synthetic / virtual sample generation	Synthesize domain diverse CSI via generative models, physically grounded perturbations, MixUp, or subcarrier dropout to broaden coverage	Adds complexity and possible distribution mismatch; DATTA instead favours lightweight augmentation for real time use.
Transfer / few shot learning	Adapt to new domains with limited labeled target data	Requires labeled target samples per new domain, which deployment rarely provides; motivates label free test time adaptation.
Test time adaptation (TTA)	Adapt a pretrained model on the unlabeled test stream by aligning feature statistics; corrects residual shift online [5]	Largely unexplored for WiFi HAR; unconstrained adaptation can drift and forget learned invariance, motivating a stability mechanism.

The progression is coherent: domain invariant methods reduce but do not eliminate shift; TTA can correct residual shift but unconstrained adaptation drifts and forgets; DATTA combines domain invariant learning with TTA and introduces random weight resetting to control forgetting [1].

6. Research Gap

Our project targets an experimental and analytical gap, not a new algorithm. Reported cross domain WiFi HAR results are often produced under favourable conditions, and the interaction among augmentation, domain adversarial learning, test time adaptation, and forgetting control is subtle and rarely isolated by independent parties. Concretely, we ask: how much improvement comes from CSI augmentation alone; how much from domain adversarial learning; how much TTA contributes independently; how important random resetting is; whether the combined method outperforms its components consistently; how sensitive performance is to the degree of domain shift; what the robustness versus computational cost trade off is; whether the published DATTA behaviour can be reproduced independently; and under what conditions the method degrades. Answering these through controlled evaluation is the project's contribution.

7. DATTA: Technical Approach

The DATTA paper proposes a framework that integrates domain adversarial training (DAT) with test time adaptation (TTA) on a lightweight backbone, with random weight resetting to preserve stability during adaptation [1]. Figure 1 shows the pipeline; the components and their roles are as follows.

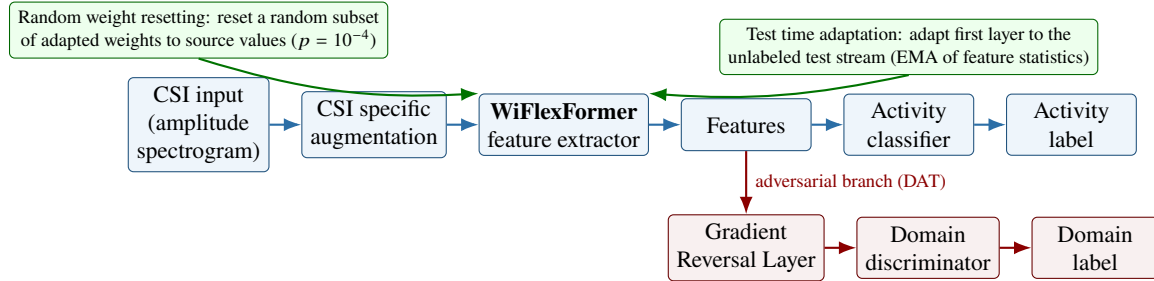


Figure 1: The DATTA pipeline (after [1]). Training path (blue): CSI amplitude input → CSI specific augmentation → WiFlexFormer feature extractor → activity classifier. Adversarial branch (red): features → Gradient Reversal Layer → domain discriminator. At inference (green): the first layer adapts to the unlabeled test stream via feature statistic alignment, and random weight resetting periodically restores a random subset of adapted weights to the source values.

Augmentation → variability. Raw amplitude spectrograms pass through a CSI specific augmentation module (amplitude perturbation, circular temporal rotation, pixel/row level operations) that increases data variability; the authors report it is essential for the adversarial component to learn invariant features [1].

WiFlexFormer feature extractor. A lightweight transformer backbone (a convolutional stem, Gaussian positional encoding, and attention) chosen for low latency, real time inference [1].

DAT → domain invariant features. A Gradient Reversal Layer connects the feature extractor to a domain discriminator; reversing the discriminator’s gradient drives the extractor toward features indistinguishable across domains, while the activity classifier preserves task information [3]. The authors use loss weights $\alpha = 0.3$, $\beta = 0.2$ and GRL scaling $\gamma = 8$, with a Confidence Control Constraint (weight $\beta = 0.2$) that discourages over confident, class imbalanced predictions [1].

TTA → residual target domain shift. At inference the model adapts its first layer to the incoming unlabeled stream by aligning feature statistics (means and variances updated via exponential moving averages), correcting residual shift without labels [1].

Random weight resetting → reduced forgetting. Each iteration, a random subset of adapted weights (a Bernoulli mask with reset rate $p = 1 \times 10^{-4}$) is reset to the source model’s parameters, mitigating catastrophic forgetting during prolonged adaptation [1].

Reported results (attributed to [1]). On Widar3.0-G6D, averaged over three runs, the DATTA authors report the components compound: the vanilla WiFlexFormer baseline reaches $\approx 40.6\%$ F1; augmentation raises it; domain adversarial training with augmentation and the Confidence Control Constraint reaches $\approx 65.7\%$ F1; and the full DATTA model reaches $\approx 68.1\%$ F1 once random weight resetting is enabled, while remaining real time. These are the authors’ results, not our team’s.

8. Widar3.0-G6D Dataset

Widar3.0 → G6 → G6D. Widar3.0 [2] is a CSI gesture dataset of 22 hand gestures performed by 16 participants across three indoor environments. Because not all gestures appear uniformly in every environment, DATTA adopts the G6 subset six gestures (push and pull, sweep, clap, slide, draw circle, draw zigzag) recorded consistently across all environments by all users and prepares it for domain

adversarial and test time evaluation; we refer to this prepared benchmark as Widar3.0-G6D [1]. The G6 subset yields approximately 68k gesture samples.

Acquisition setup. A single transmitter and six receivers, each with an Intel WiFi Link 5300 at 5 GHz, collect CSI from 3 antennas $\times$ 30 subcarriers at 1,000 Hz using the Linux CSI Tool [1].

Temporal processing and normalization. DATTA subsamples each sequence to 120–220 packets (≈ 1.2 – 2.2 s), zero padding shorter sequences, and min max normalizes amplitude features. The prepared dataset comprises approximately 58k samples [1].

Two sample counts, two stages. The $\approx 68k$ figure is the raw G6 gesture sample count; the $\approx 58k$ figure is the prepared sample count after DATTA’s temporal subsampling and filtering. They describe different stages of the same pipeline and are not contradictory [1].

Domain definition and split. A domain is a unique room participant (environment user) combination. The prepared data is partitioned by domain into disjoint subsets so that test domains are unseen during training: per [1], training uses 7 domains (2 environments, 7 users) and testing uses the remaining 9 domains (1 environment, 9 users). Because the split is by whole domains rather than by random samples, it measures genuine cross domain generalization rather than within domain memorization, which is what makes the benchmark appropriate for this project. The authors release a script that regenerates these exact subsets [1].

9. Experimental Protocol: Train / Val / Val_TTA / Test

DATTA uses four subsets with distinct roles [1]. Train fits the model backbone. Val is the ordinary source model validation/selection set used during standard training. Val_TTA is a separate set used only to tune the test time adaptation settings (for example the adaptation rate and reset rate). Test is the final evaluation set, evaluated last.

Why Val_TTA is separated from Test. If TTA settings were tuned on Test, the adaptation configuration would be fitted to the evaluation distribution and reported robustness would be inflated; tuning them on Val would conflate source model selection with adaptation tuning. Isolating Val_TTA keeps three concerns separate source model selection (Val), TTA schedule tuning (Val_TTA), and unbiased final evaluation (Test) which prevents leakage and preserves experimental validity. DATTA additionally evaluates on a shuffled Test to simulate an extreme case of high frequency domain shifts [1].

Secondary datasets (context only). The DATTA paper reports additional cross dataset generalization on 3DO (a through wall single participant dataset) and MM-Fi (a multimodal dataset) [1]; these are the authors’ generalization tests and are not our project’s primary benchmark. Our implementation and evaluation target Widar3.0-G6D.

10. Current Project Technical Direction

Our team will implement and evaluate the DATTA pipeline on Widar3.0-G6D in stages of increasing completeness, mirroring the paper’s configuration names [1]: (i) reproduce the vanilla WiFlexFormer baseline (W) and measure the cross domain gap; (ii) add the CSI augmentation module; (iii) add domain adversarial training (Gradient Reversal Layer, domain discriminator, activity classifier, and the Confidence Control Constraint), giving W_DAT; (iv) add test time adaptation, giving W_TTA and W_DATTA; (v) add random weight resetting. At each stage we will measure accuracy and macro F1 on unseen test domains and compare against the previous stage. Where compute does not permit the full three run, all domain protocol, we will reduce the number of domains or runs while preserving the split discipline and configuration definitions.

11. SOTA Positioning

Within the literature and benchmark setting reviewed for this project, DATTA is the most directly relevant recent reference for real time cross domain WiFi HAR [1]. It is the key reference because it targets the cross domain objective explicitly; combines domain adversarial training, test time adaptation, and random weight resetting; uses a lightweight WiFlexFormer backbone suited to real time operation; is evaluated on Widar3.0-G6D; and reports comparisons against a domain adversarial baseline (DAT) [4] and a test time adaptation baseline (ViTTA) [5], with improvements of up to 8.1 F1 points over the strongest of these within the benchmark setting studied [1].

Table 2: SOTA reference points and our relationship to them.

Reference point	Represents	Our relationship
DAT baseline [4]	Domain adversarial WiFi HAR baseline used in [1]	Reproduced as the intermediate configuration W_DAT
ViTTA [5]	Test time adaptation baseline used in [1]	Comparison point for the TTA component
DATTA (full) [1]	The most relevant reference method on Widar3.0-G6D	Technical foundation reproduced, evaluated, and analyzed

Contribution boundary. We do not claim our project is state of the art, nor that DATTA is the global state of the art beyond the setting reviewed here. Our contribution is an independent reproduction, a controlled component wise evaluation, a reproducibility assessment, and a robustness versus complexity analysis. Any results we later report will be clearly our own and separate from the paper's.

12. M1 Technical Progress / Current Status

At Milestone 1 our team has fixed the technical direction (DATTA) and the primary dataset (Widar3.0-G6D); analyzed the method's components and the dataset's structure, including the Train/Val/Val_TTA/Test protocol and the domain based split; identified the authors' released code and subset preparation script as the reproduction starting point; and defined the staged plan in Sections 10 and 15. No model has been trained and no result has been produced by our team. Consistent with the Milestone 1 objectives (problem definition, review of existing work, SOTA positioning, and timeline), this report deliberately contains zero team generated experimental results; every quantitative figure herein is attributed to the DATTA paper [1].

13. Timeline / Work Plan

Table 3: Planned technical work by milestone.

Milestone	Planned technical work
M1 by 13 Sep	Problem definition, literature review, SOTA positioning, dataset and experimental protocol scoping, staged plan (this report); repository initialized
M2 by 11 Oct	Acquire and prepare Widar3.0-G6D with the authors' script; understand and preprocess CSI (subsampling, min max normalization); construct the domain based Train/Val/Val_TTA/Test split; reproduce the WiFlexFormer baseline (W) and measure the cross domain gap
M3 by 1 Nov	Add the augmentation module; implement DAT (GRL + domain discriminator + activity classifier + Confidence Control Constraint); implement TTA; implement random weight resetting; run the component wise ablation and compare against the DAT and ViTTA baselines; produce benchmark tables and plots
M4 by 22 Nov	Full evaluation and robustness/complexity analysis (accuracy, macro F1, generalization gap, parameter count, inference time); final IEEE format report, refined SOTA positioning, reproducible repository, video, and demo

14. Expected Technical Evaluation

We will evaluate the configurations W, W with augmentation, W_DAT, W_TTA, and W_DATTA (with and without random weight resetting), reproducing the paper's ablation structure [1]. Reported measures will be classification accuracy and macro averaged F1 on unseen test domains; the cross domain generalization gap between in domain and unseen domain performance; and the accuracy complexity trade off via parameter count and per sample inference time. Splits, seeds, and metrics will be fixed and documented. We will not report any measurement our team has not actually obtained.

15. Limitations and Risks

Table 4: Project risks and planned mitigations.

Risk	Mitigation
Compute: full three run, all domain training with DAT and TTA is expensive	Reduce domains/runs and use free tier GPU; preserve the design rather than shrink it silently
Reproduction fidelity: exact hyperparameters/preprocessing may be under specified	Follow the released code and script; document any deviation explicitly
Dataset preparation must match the authors' subsets to stay comparable	Regenerate subsets with the authors' script; do not hand alter the split
Dependence on released code	Verify behaviour on a small subset before scaling; record versions/commits
TTA computational overhead at inference	Measure inference time explicitly; report the accuracy cost trade off honestly
Hyperparameter sensitivity (α , β , γ , reset rate p)	Tune only on Val_TTA; report sensitivity rather than cherry picking
Published vs reproduced results may differ	Report our numbers as our own; analyze and document discrepancies instead of tuning to match
Exact hardware/runtime conditions cannot be reproduced	State our hardware; compare trends, not absolute latencies, where hardware differs

16. Conclusion

This report establishes the problem (cross domain WiFi CSI HAR), a literature progression that motivates the chosen foundation, the technical approach (DATTA: domain adversarial training with test time adaptation and random weight resetting on a lightweight WiFlexFormer backbone), the primary dataset (Widar3.0-G6D, with its domain based split and Train/Val/Val_TTA/Test protocol), a qualified state of the art position, and a staged plan to reproduce, evaluate, and analyze the method. It demonstrates understanding of the problem, the relevant reference method, the dataset, and the experimental protocol, and commits to reporting our own results separately and honestly at later milestones.

References

- [1] J. Strohmayer, R. Sterzinger, M. Wodlinger, and M. Kampel, "DATTA: Domain-Adversarial Test-Time Adaptation for Cross-Domain WiFi-Based Human Activity Recognition," IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026. Code: <https://github.com/StrohmayerJ/DATTA>.
- [2] Y. Zheng et al., "Widar3.0: Zero-Effort Cross-Domain Gesture Recognition with Wi-Fi," as cited in [1] (source of the Widar3.0 dataset and its G6 subset).
- [3] Y. Ganin and V. Lempitsky, "Unsupervised Domain Adaptation by Backpropagation," (Domain-Adversarial Neural Network / Gradient Reversal Layer), as cited in [1].
- [4] W. Jiang et al., "Towards Environment-Independent Device-Free Human Activity Recognition," (domain adversarial WiFi HAR baseline), as cited in [1].
- [5] W. Lin et al., "Video Test-Time Adaptation (ViTTA)," (test time adaptation baseline), as cited in [1].